

Topic-Guided Multi-View Deep Wasserstein Clustering for Single-cell Transcriptomic Analysis

I. IMPLEMENTATION DETAILS ON THE 'MRCA_BC' DATASET

We filtered genes and cells using Scanpy [1]: applying 'sc.pp.filter_genes' with min_counts=10 and 'sc.pp.filter_cells' with min_counts=100. For input features to the encoders of the multi-view cell transcriptional encoder, we selected the top 1000 and 500 highly variable genes. Additionally, we applied the $\log(1 + \mathbf{X})$ transformation to $\mathbf{X}$ to derive $\mathbf{X}_n$, with 1000 highly variable genes, as the other input of the encoder. Notably, batch training was utilized. For pre-training and fine-tuning stages, the learning rates were both set to $1e-5$, with 256 batches. Pre-training and fine-tuning were conducted for 50 and 20 epochs, respectively. The multi-view encoder comprised three neural networks with single-layer dimensions of 256, 256, and 128 nodes. The topic-guided ZINB decoder featured a 512-node layer, while the OT regularized topic decoder had $\lambda_z = 1.0$ as the MMD weight parameter. The latent dimension was set to 20. Other implementation details of baselines are provided in Table II.

II. BASELINE METHODS

The baseline methods include deep embedded methods (scDeepCluster [2], scziDesk [3], scDCC [4], DeepScena [5], scMAE [6]), deep graph embedded methods (GraphSCC [7], scTAG [8], scGPCL [9], scLEGA [10], scHNTL [11]), and two traditional top non-neural network-based approaches (SC3 [12], Seurat [13]). The details of these methods are as follows:

- **scDeepCluster**(2019) [2]: scDeepCluster is a model-based deep embedded clustering method designed for scRNA-seq data.
- **scziDesk**(2020) [3]: scziDesk proposed deep soft K-means, which enhances performance by utilizing a self-training process.
- **scDCC**(2021) [4]: A single-cell deep constrained clustering method that incorporates prior knowledge to guide clustering.
- **DeepScena**(2023) [5]: DeepScena combines a NB-based convolutional autoencoder with self-supervised similarity learning to improve clustering, especially for large-scale data and rare cell types.
- **scMAE**(2024) [6]: scMAE applies a masked autoencoder with a masking predictor to capture gene dependencies and learn robust embeddings for clustering.
- **GraphSCC**(2020) [7]: GraphSCC leverages graph convolutional networks to consider cellular structural relationships for improved clustering.
- **scTAG**(2022) [8]: scTAG is the first graph embedding autoencoder method for scRNA-seq data which uses a ZINB model to learn the distribution of data.

- **scGPCL**(2023) [9]: scGPCL uses GNNs on cell-gene graphs with prototypical contrastive learning to preserve relational information and improve robustness to dropout.
- **scLEGA**(2024) [10]: scLEGA integrates a low-expression optimized denoising AE and a GCN-based graph AE, guided by multi-head attention and ZINB loss, for accurate clustering.
- **scHNTL**(2025) [11]: scHNTL exploits high-order neighbors and triplet loss in a graph attentional autoencoder to enhance separability of cell clusters.
- **SC3**(2017) [12]: A consensus clustering tool for scRNA-seq data that integrates multiple algorithms to provide robust and accurate cluster assignments.
- **Seurat**(2018) [13]: An R package for scRNA-seq data that offers comprehensive analytical capabilities.

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TABLE I
SUMMARY OF NOTATIONS

Variable	Definition
N	the number of cells
M	the number of genes
L	the dimensions of latent space
U	the number of the grand truth
K	the number of the views in the multi-view encoder
$\mathbf{X}$	the raw data of the single cell RNA-seq
$\mathbf{X}_n$	the smoothed non-negative data from $\mathbf{X}$
ϵ	the random Gaussian noise in the multi-view encoder
$\mathbf{X}_{input_i}$	the input of the encoder _{i} without a random Gaussian noise
$\mathbf{X}_i$	the input of the encoder _{i} with a random Gaussian noise
$\mathcal{W}_i^{(\ell)}, b_i^{(\ell)}$	learnable weights in the multi-view encoder
$\mathbf{Z}_i^{(\ell)}$	the output of the ℓ -th layer of encoder _{i}
$\phi(\cdot)$	the activation function in the multi-view encoder
$\mathbf{Z}_i$	the cell embedding of the encoder _{i}
k_i	the weight assigned to the i -th cell embedding in the multi-view encoder
$\mathbf{Z}$	the cell embedding in the latent space
$G(\mathbf{X}, \mathbf{E})$	cell graph with input $\mathbf{X}$ and weight matrix $\mathbf{E}$
$\exp(\cdot)$	the exponential function
σ	the variance scale parameter in heat kernel function
$\mathbf{W}$	the cell-topic embedding
$\mathbf{H}$	the topic-gene embedding
$\mathbf{A}$	the cell graph matrix in topic model
T	the order of cell graph
$\mathbf{D}$	the degree matrix of cell graph in topic model
$\mathbf{L}$	the graph Laplacian matrix in topic model
$\mathcal{R}_{GS}$	the graph-sparse regularization term in topic model
λ_1, λ_2	the weight parameters of the graph regularization and sparse regularization in topic model, respectively
$\mathcal{L}_{GS}$	the objective function of topic model
Φ, Ψ	the Lagrange multiplier for cell-topic embedding $\mathbf{W}$ and topic-gene embedding $\mathbf{H}$, respectively
$\mathcal{J}_{GS}$	the Lagrangian function in topic model
$\mathbf{I}$	the identity matrix
$Tr(\cdot)$	the trace of a matrix
$\odot$	the element-wise multiplication
$\sigma(\cdot)$	the sparse transformation function
$\mathbf{V}$	the output of the sparse transformation function
$\mathbf{S}$	the input of the topic-guided ZINB decoder
λ_w	the topic influence coefficient
$\text{sigmoid}(\cdot)$	the logistic sigmoid function
f_D	the decoder function
$\mathbf{W}_\pi, \mathbf{W}_\theta, \mathbf{W}_\mu$	the learned weights of the three fully connected layers in topic-guided ZINB decoder
$\text{NB}(\cdot)$	the Negative Binomial model
$\text{ZINB}(\cdot)$	the Zero-Inflated Negative Binomial model
$\mathcal{L}_{zinb}$	the loss function of topic-guided ZINB decoder
$\Pi, \Theta, \mathbf{M}$	the parameter matrices of ZINB model
$\tilde{\mathbf{X}}$	the reconstructed data of the topic decoder
$\mathcal{L}_{topic}$	the loss function of topic decoder
$P_{\mathbf{X}}, P_{\mathbf{Y}}$	the original data distribution and the topic decoder reconstruction distribution, respectively.
$c(\cdot, \cdot)$	the measurable cost function in OT loss function
$\mathcal{P}(\mathbf{X} \sim P_{\mathbf{X}}, \mathbf{Y} \sim P_{\mathbf{Y}})$	a set of all joint distributions of $(\mathbf{X}, \mathbf{Y})$ with marginals $P_{\mathbf{X}}$ and $P_{\mathbf{Y}}$ respectively
$G(\mathbf{Z})$	the topic decoder with the cell embedding $\mathbf{Z}$ as input
$\mathcal{Q}$	a set of multi-view cell transcriptional encoders
$Q(\mathbf{Z} \mathbf{X})$	a multi-view encoder in $\mathcal{Q}$ with input $\mathbf{X}$ and output $\mathbf{Z}$
$Q(\mathbf{Z})$	the cell embedding distribution in the latent space
$P(\mathbf{Z})$	the standard Gaussian prior distribution
$\mathcal{L}_{mmd}$	the MMD loss function
$k(\cdot, \cdot)$	the kernel function in OT regularized topic decoder
$\mathcal{L}_{ot}$	the loss function of OT regularized topic decoder
λ_z	the weight coefficient of MMD loss
$Q_{dec_{iu}}$	the soft assignment distribution between i -th cell and u -th cluster center
$P_{dec_{iu}}$	the auxiliary target distribution between i -th cell and u -th cluster center
$\text{KL}(\cdot \cdot)$	the Kullback-Leibler divergence between two distributions
$\mathcal{L}_{cluster}$	the loss function of the clustering stage
$\mathcal{L}_{total}$	the total loss of scMTO
$\gamma_1, \gamma_2, \gamma_3$	the weight coefficients of $\mathcal{L}_{zinb}$, $\mathcal{L}_{ot}$ and $\mathcal{L}_{cluster}$
$\mathbf{Y}_i$	the predict label of the i -th cell

TABLE II
PERFORMANCE OF DIFFERENT MODELS ACROSS PRE-TRAIN AND FINE-TUNE EPOCHS ON 'MRCA_BC' DATASET.

	NMI	ARI	Pre-train(epochs)	Fune-train(epochs)	HVGs
scMTO	0.9151	0.8457	50	20	1000, 500
scDeepCluster	0.8311	0.7543	default	default	1000
scziDesk	0.6052	0.3926	default	default	default
scDCC	0.8434	0.7684	default	default	1000
DeepScena	0.5671	0.3786	default	default	default
scMAE	0.8488	0.7986	default	default	default
GraphSCC	0.0747	0.018	default	default	1000

TABLE III
THE CLUSTERING PERFORMANCE OF SCMTO WITH DIFFERENT λ_z .

	Datasets	$\lambda_z=0$	$\lambda_z=0.1$	$\lambda_z=0.3$	$\lambda_z=0.5$	$\lambda_z=0.7$	$\lambda_z=0.9$	$\lambda_z=1.0$
NMI	Pollen	0.9488	0.9496	0.9455	0.9548	0.9466	0.9456	0.941
	Camp_Liver	0.8495	0.8476	0.8476	0.8462	0.8439	0.8396	0.8463
	QS_Diaphragm	0.9588	0.9588	0.9588	0.9624	0.9624	0.9624	0.9624
	QS_Limb_Muscle	0.9451	0.9497	0.9464	0.9484	0.9459	0.9659	0.9459
	QS_Lung	0.8089	0.8162	0.8193	0.8172	0.8274	0.8472	0.8455
	Muraro	0.8947	0.8962	0.8966	0.8894	0.8928	0.8922	0.8909
	Romanov	0.7418	0.7415	0.7403	0.7385	0.7266	0.7409	0.7339
	Adam	0.8127	0.8115	0.8188	0.8959	0.8937	0.887	0.8896
	QS_Heart	0.9069	0.9064	0.9034	0.8876	0.9021	0.8828	0.8814
	Young	0.7529	0.8239	0.7567	0.7551	0.8224	0.8218	0.8215
	Wang_Lung	0.9091	0.9091	0.9173	0.9201	0.9174	0.9211	0.923
	Qx_Spleen	0.8674	0.8658	0.8723	0.8775	0.8795	0.8782	0.8773
	Average	0.8664	0.873	0.8686	0.8744	0.8801	0.8821	0.8799
ARI	Pollen	0.9506	0.9545	0.9564	0.9547	0.9508	0.9529	0.9486
	Camp_Liver	0.735	0.7356	0.7356	0.7339	0.7316	0.7276	0.7324
	QS_Diaphragm	0.9754	0.9754	0.9754	0.9783	0.9783	0.9783	0.9783
	QS_Limb_Muscle	0.9635	0.9665	0.9661	0.9682	0.9702	0.9802	0.9645
	QS_Lung	0.7443	0.7471	0.7431	0.7495	0.76	0.8446	0.843
	Muraro	0.9343	0.9356	0.9356	0.9314	0.9339	0.9337	0.9335
	Romanov	0.7882	0.7831	0.783	0.7787	0.7627	0.777	0.7705
	Adam	0.7785	0.7763	0.7851	0.9116	0.9075	0.9005	0.9038
	QS_Heart	0.9427	0.9417	0.9388	0.9448	0.9388	0.9427	0.9411
	Young	0.5767	0.7202	0.5773	0.5771	0.7236	0.7181	0.7085
	Wang_Lung	0.9621	0.9621	0.9662	0.9676	0.9662	0.968	0.969
	Qx_Spleen	0.9359	0.933	0.9381	0.941	0.9445	0.9427	0.9416
	Average	0.8573	0.8693	0.8584	0.8697	0.8807	0.8889	0.8862

TABLE IV
THE CLUSTERING PERFORMANCE OF SCMTO WITH DIFFERENT λ_w .

	Datasets	$\lambda_w=0$	$\lambda_w=0.1$	$\lambda_w=0.3$	$\lambda_w=0.5$	$\lambda_w=0.7$	$\lambda_w=0.9$	$\lambda_w=1$
NMI	Pollen	0.9411	0.9456	0.9331	0.9107	0.8538	0.8594	0.8942
	Camp_Liver	0.8398	0.8396	0.8469	0.7793	0.7881	0.7203	0.6327
	QS_Diaphragm	0.9665	0.9624	0.9543	0.9413	0.9311	0.918	0.9248
	QS_Limb_Muscle	0.9603	0.9659	0.9448	0.9472	0.9439	0.9325	0.8829
	QS_Lung	0.8169	0.8472	0.8345	0.8509	0.8334	0.8292	0.8342
	Muraro	0.8972	0.8922	0.8462	0.8363	0.8326	0.8322	0.827
	Romanov	0.7377	0.7409	0.7042	0.6894	0.6692	0.626	0.6256
	Adam	0.8023	0.887	0.8179	0.8178	0.8123	0.8036	0.8068
	QS_Heart	0.8767	0.8828	0.9011	0.8817	0.8854	0.8476	0.8484
	Young	0.798	0.8218	0.8114	0.7481	0.7566	0.7149	0.7138
	Wang_Lung	0.9221	0.9211	0.9221	0.9222	0.9241	0.925	0.9259
	Qx_Spleen	0.8734	0.8782	0.724	0.6722	0.6561	0.6274	0.6441
	Average	0.8693	0.8821	0.8534	0.8331	0.8239	0.803	0.7967
ARI	Pollen	0.9485	0.9529	0.9373	0.9133	0.7661	0.8538	0.8751
	Camp_Liver	0.7269	0.7276	0.7342	0.6691	0.6681	0.5335	0.369
	QS_Diaphragm	0.9804	0.9783	0.9747	0.9654	0.9594	0.9495	0.9553
	QS_Limb_Muscle	0.9773	0.9802	0.9692	0.9718	0.9673	0.9602	0.9341
	QS_Lung	0.7464	0.8446	0.8369	0.8476	0.8367	0.8301	0.831
	Muraro	0.9364	0.9337	0.8881	0.8778	0.8759	0.868	0.8669
	Romanov	0.7912	0.777	0.7244	0.6989	0.6596	0.5888	0.5904
	Adam	0.7634	0.9005	0.7383	0.7413	0.7405	0.7354	0.7411
	QS_Heart	0.9368	0.9427	0.9362	0.9383	0.9277	0.8763	0.8764
	Young	0.6828	0.7181	0.7115	0.671	0.6884	0.5706	0.5696
	Wang_Lung	0.9685	0.968	0.9685	0.9685	0.9694	0.9699	0.9703
	Qx_Spleen	0.9434	0.9427	0.8264	0.7834	0.7594	0.7028	0.7268
	Average	0.8668	0.8889	0.8538	0.8372	0.8182	0.7866	0.7755

TABLE V
THE CLUSTERING PERFORMANCE OF SCMTO WITH DIFFERENT γ_1 .

	Datasets	$\gamma_1=0.1$	$\gamma_1=0.2$	$\gamma_1=0.3$	$\gamma_1=0.4$	$\gamma_1=0.5$	$\gamma_1=0.7$	$\gamma_1=0.9$	$\gamma_1=0.91$	$\gamma_1=0.93$	$\gamma_1=0.95$	$\gamma_1=0.97$	$\gamma_1=0.99$	$\gamma_1=1$
NMI	Pollen	0.7599	0.6836	0.7067	0.7665	0.7062	0.6783	0.8468	0.8566	0.9326	0.9269	0.9358	0.9456	0.9385
	Camp_Liver	0.7424	0.7368	0.7353	0.7433	0.7383	0.6461	0.6616	0.6613	0.7313	0.7311	0.7452	0.8396	0.8246
	QS_Diaphragm	0.9035	0.9095	0.9193	0.7913	0.7959	0.8207	0.854	0.854	0.8661	0.8941	0.9211	0.9624	0.9458
	QS_Limb_Muscle	0.7998	0.7998	0.7998	0.8045	0.8032	0.81	0.9137	0.9155	0.914	0.9203	0.9312	0.9659	0.9521
	QS_Lung	0.7271	0.7279	0.7257	0.7268	0.7268	0.7413	0.7476	0.7485	0.7507	0.8329	0.8422	0.8472	0.7621
	Muraro	0.8436	0.8478	0.8477	0.8503	0.8501	0.8506	0.8538	0.8486	0.8441	0.845	0.8412	0.8922	0.892
	Romanov	0.6013	0.6277	0.6223	0.6625	0.6607	0.6905	0.6145	0.6282	0.6252	0.6425	0.688	0.7409	0.7458
	Adam	0.7303	0.7301	0.7292	0.7295	0.726	0.7011	0.852	0.8023	0.8556	0.868	0.8799	0.887	0.9051
	QS_Heart	0.883	0.8861	0.8867	0.8886	0.8878	0.8891	0.8923	0.9132	0.8919	0.8893	0.8874	0.8828	0.9247
	Young	0.6193	0.5187	0.6129	0.6688	0.6462	0.5951	0.5929	0.6291	0.7043	0.7076	0.76	0.8218	0.8169
	Wang_Lung	0.0118	0.0121	0.012	0.012	0.0117	0.0122	0.0007	0.0003	0.8554	0.9095	0.9239	0.9211	0.9093
	Qx_Spleen	0.4447	0.4449	0.4559	0.4322	0.727	0.7055	0.626	0.6705	0.6322	0.6492	0.6927	0.8782	0.6272
	Average	0.6722	0.6604	0.6711	0.673	0.69	0.6784	0.7047	0.7107	0.8003	0.818	0.8374	0.8821	0.8537
ARI	Pollen	0.5014	0.3559	0.3846	0.5048	0.3788	0.3561	0.7051	0.7065	0.9375	0.9366	0.9468	0.9529	0.9428
	Camp_Liver	0.5256	0.5212	0.5179	0.5252	0.5175	0.3901	0.4083	0.4098	0.4837	0.4778	0.497	0.7276	0.7095
	QS_Diaphragm	0.9254	0.9312	0.9409	0.7853	0.7923	0.8286	0.8662	0.8662	0.8806	0.9115	0.9386	0.9783	0.9714
	QS_Limb_Muscle	0.7834	0.7834	0.7834	0.7894	0.7843	0.7923	0.9342	0.9373	0.9389	0.9428	0.9488	0.9802	0.9709
	QS_Lung	0.6379	0.6391	0.6375	0.6391	0.6391	0.6601	0.6702	0.6725	0.6784	0.8347	0.8263	0.8446	0.5805
	Muraro	0.8861	0.8899	0.8898	0.8928	0.8925	0.8919	0.8966	0.8905	0.8846	0.8851	0.8805	0.9337	0.9324
	Romanov	0.7012	0.7201	0.7086	0.7284	0.7513	0.7583	0.5889	0.6045	0.5804	0.6427	0.6803	0.777	0.769
	Adam	0.6837	0.6827	0.6757	0.6772	0.674	0.6566	0.8676	0.8108	0.8739	0.8852	0.8891	0.9005	0.9197
	QS_Heart	0.9268	0.9296	0.9301	0.9322	0.9311	0.9335	0.9456	0.9575	0.9468	0.9451	0.9449	0.9427	0.9656
	Young	0.4262	0.2971	0.4225	0.548	0.527	0.424	0.4226	0.446	0.5573	0.5555	0.6192	0.7181	0.7025
	Wang_Lung	-0.0355	-0.0358	-0.0356	-0.0355	-0.0353	-0.0361	-0.006	-0.0041	0.9231	0.96	0.9694	0.968	0.9621
	Qx_Spleen	0.3205	0.3202	0.3446	0.3266	0.8296	0.8249	0.7183	0.8236	0.7164	0.6964	0.8091	0.9427	0.5007
	Average	0.6069	0.5862	0.6	0.6095	0.6402	0.6234	0.6681	0.6768	0.7835	0.8061	0.8292	0.8889	0.8273

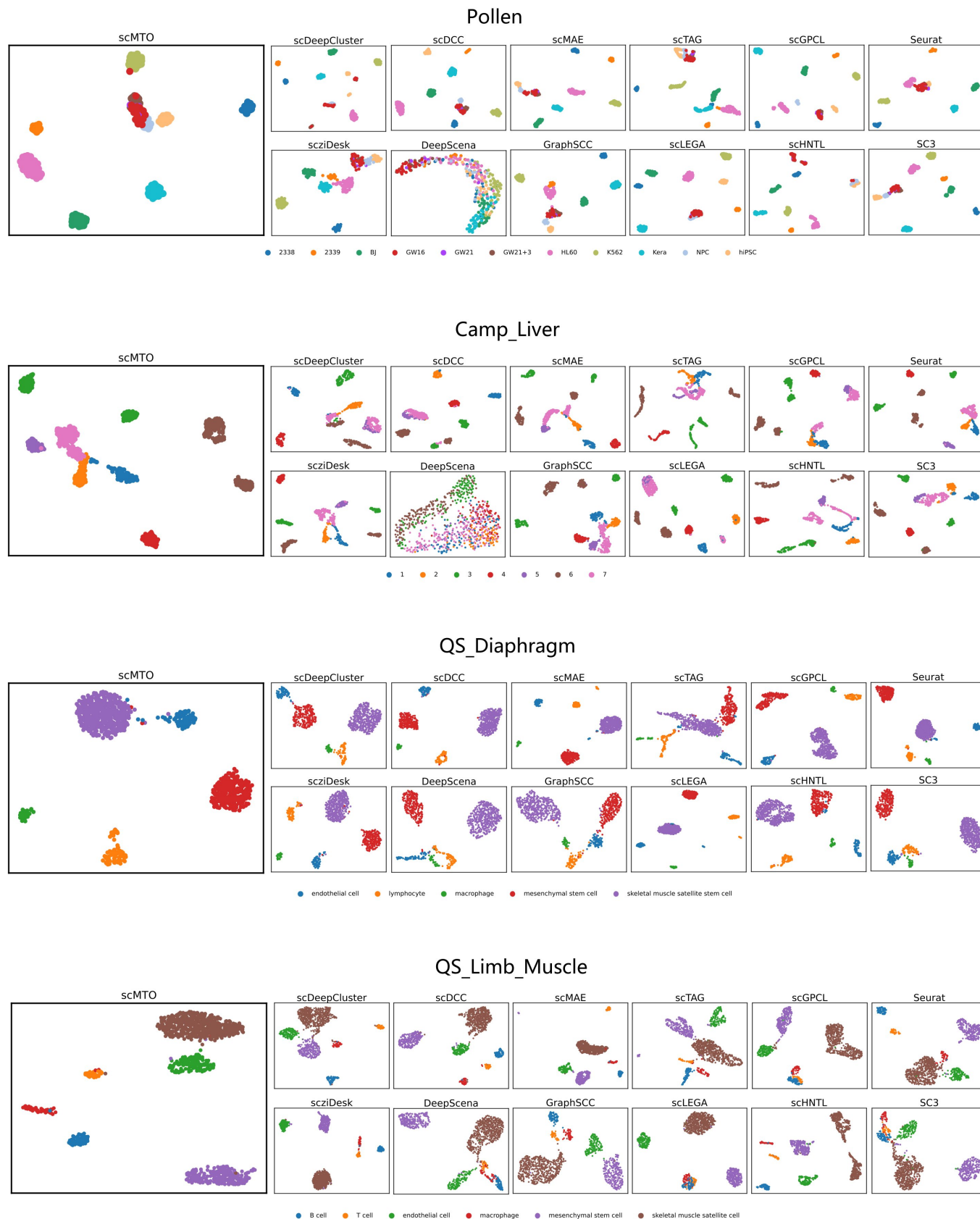


Fig. 1. Comparison of clustering results with 2D visualization by UMAP on 'Pollen', 'Camp_Liver', 'QS_Diaphragm', 'QS_Limb_Muscle' datasets.

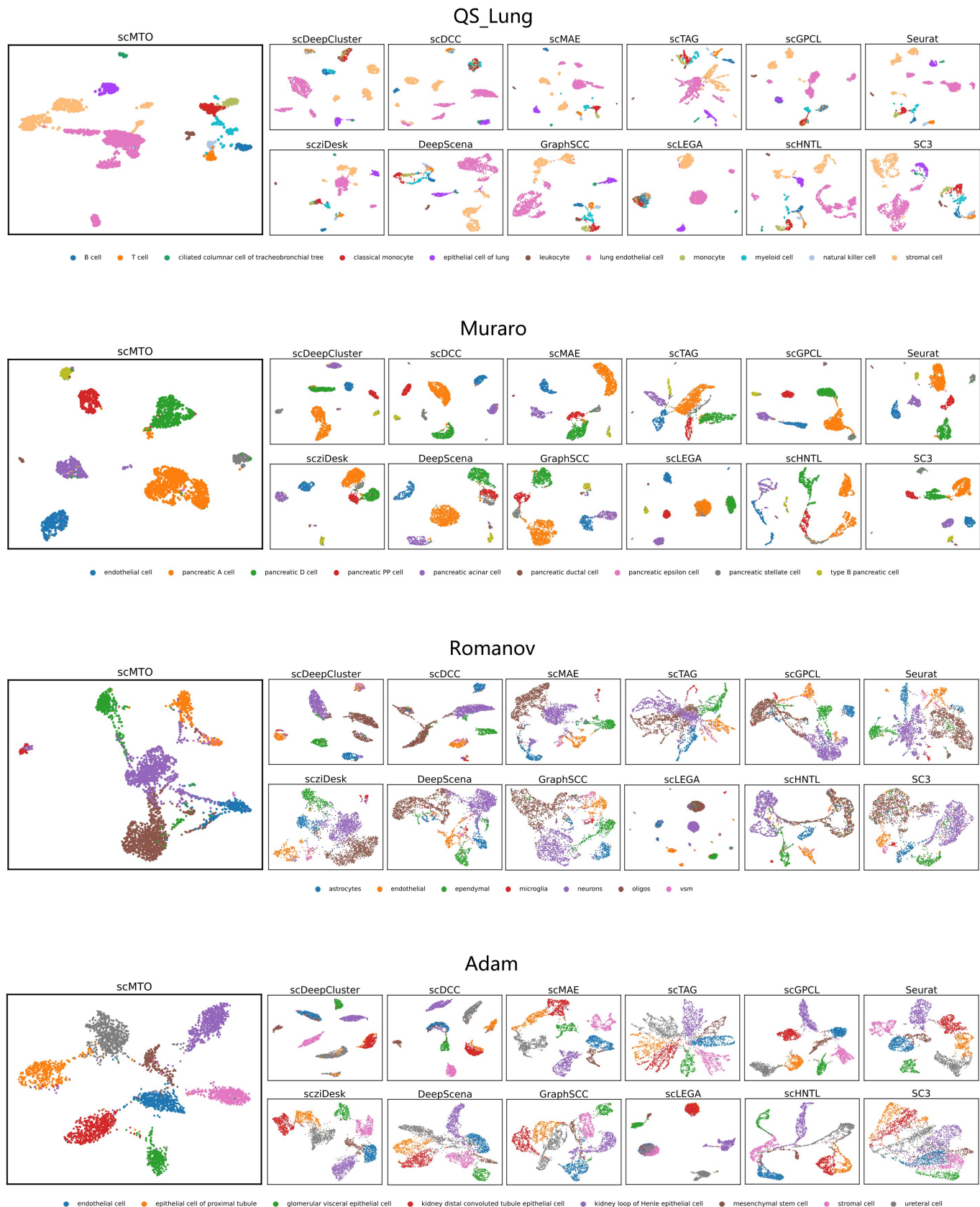


Fig. 2. Comparison of clustering results with 2D visualization by UMAP on 'QS_Lung', 'Muraro', 'Romanov', 'Adam' datasets.

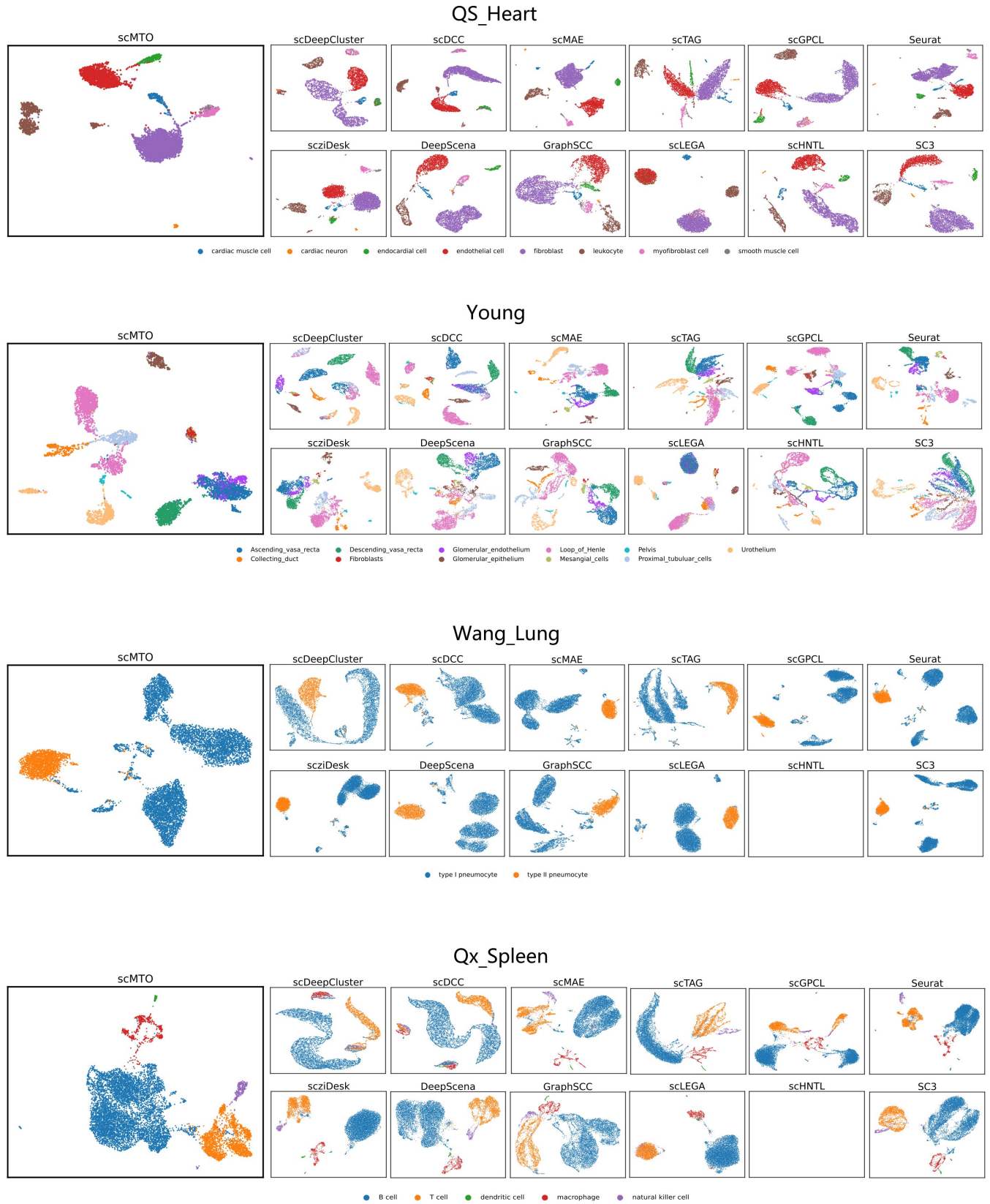


Fig. 3. Comparison of clustering results with 2D visualization by UMAP on 'QS_Heart', 'Young', 'Wang_Lung', 'Qx_Spleen' datasets.

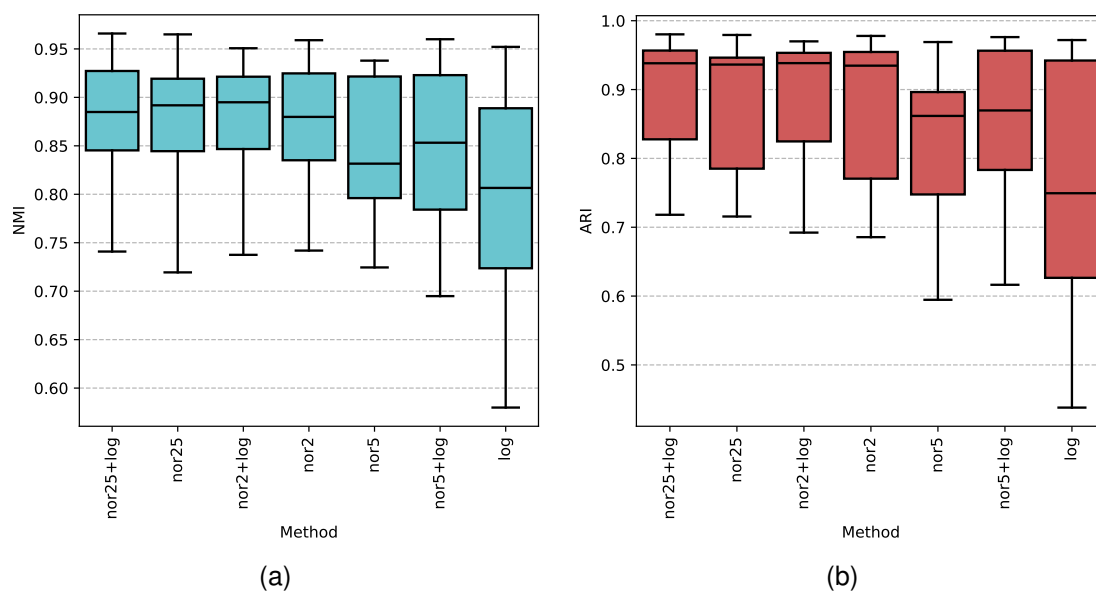


Fig. 4. The boxplot depicting the comparison of clustering results with different view selections: (a) NMI; (b) ARI.

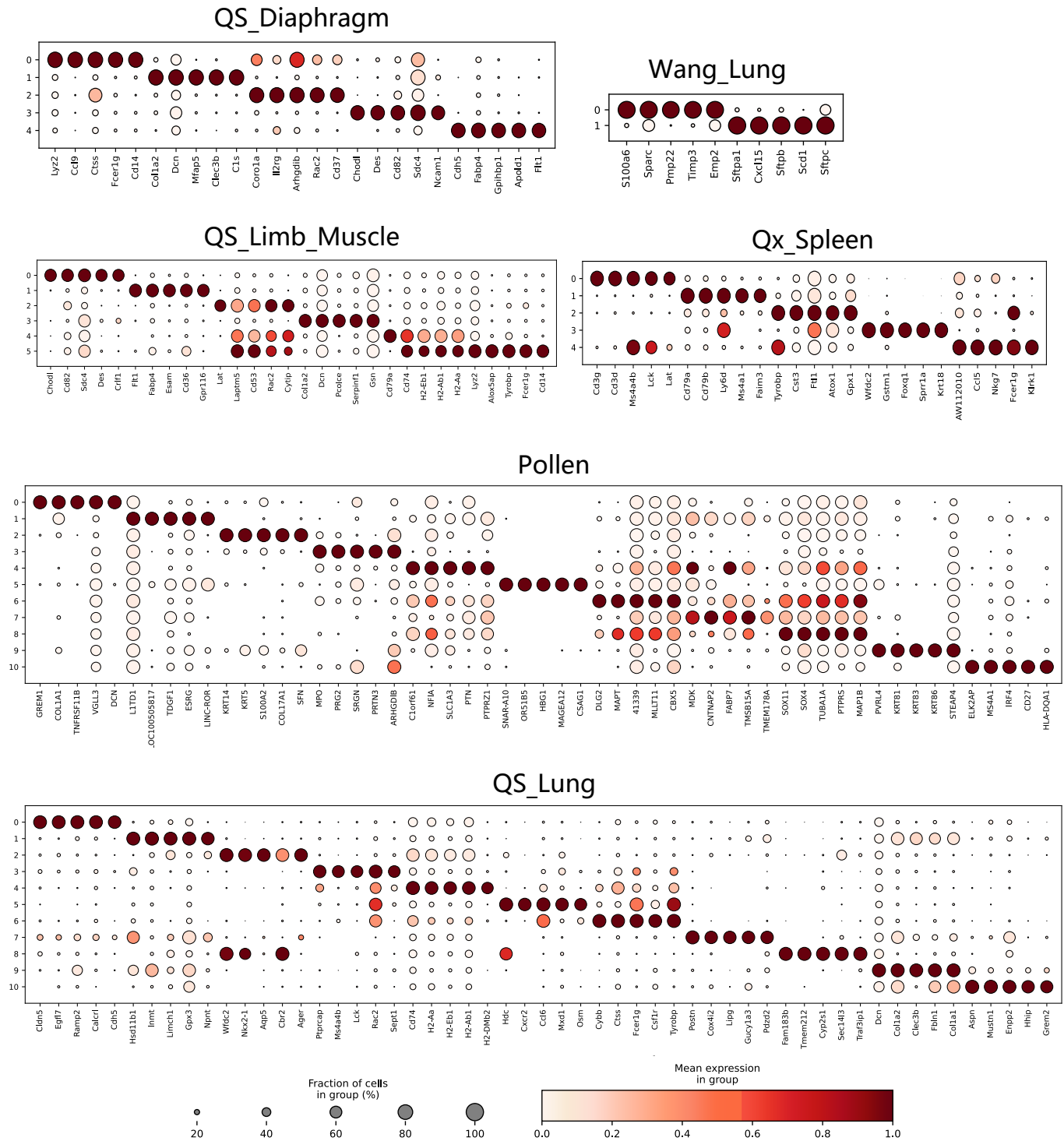


Fig. 5. Dot plots of the top 5 genes expressed in each cluster of scMTO on 'QS_Diaphragm', 'Wang_Lung', 'QS_Limb_Muscle', 'Qx_Spleen', 'Pollen', 'QS_Lung' datasets.

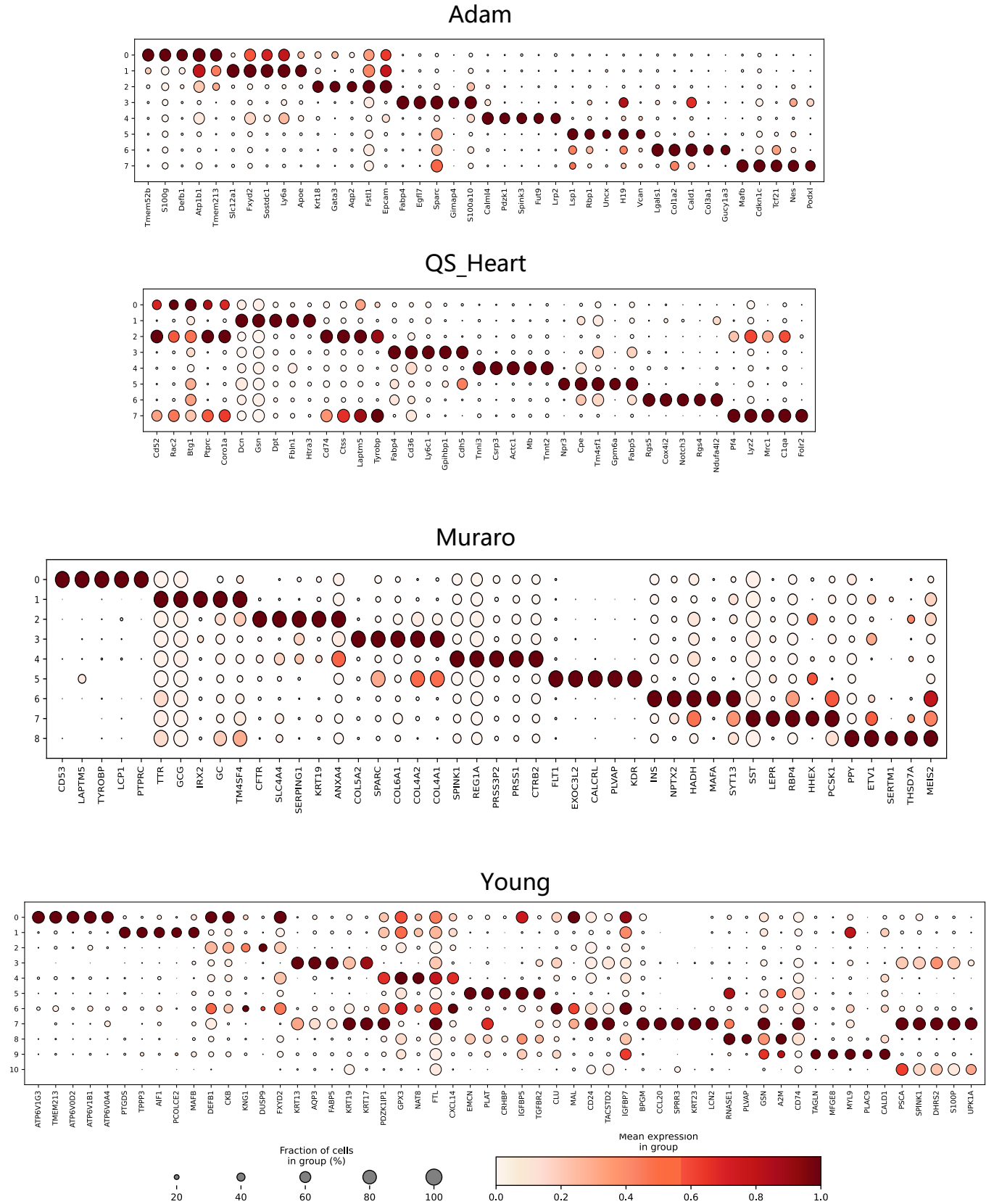


Fig. 6. Dot plots of the top 5 genes expressed in each cluster of scMTO on 'Adam', 'QS_Lung', 'Muraro', 'Young' datasets.

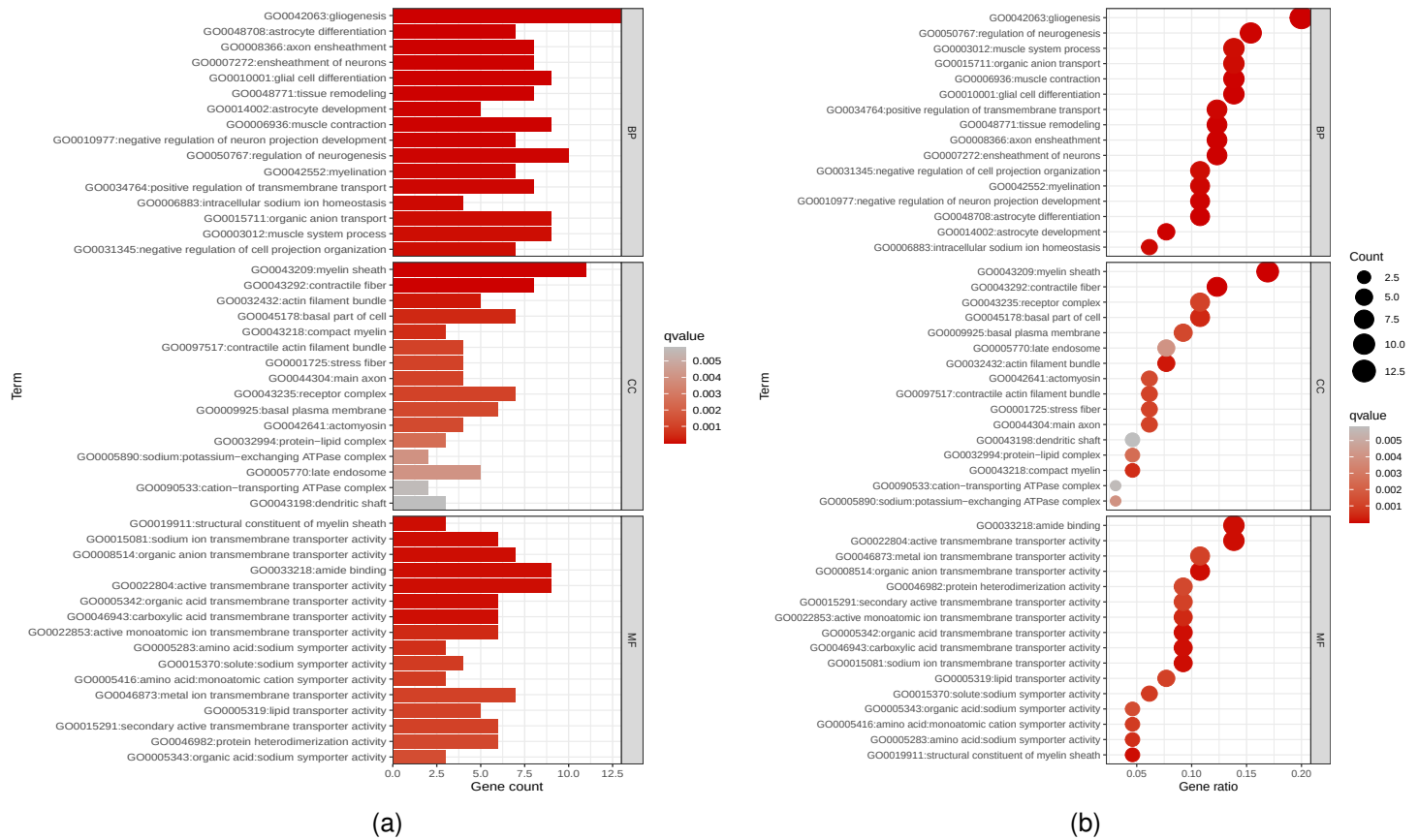


Fig. 7. GO Enrichment study on the 'Romanov' dataset: (a) bar plot; (b) bubble plot.